

Supply Chain Coordination

Instructor: Sairam Sriraman

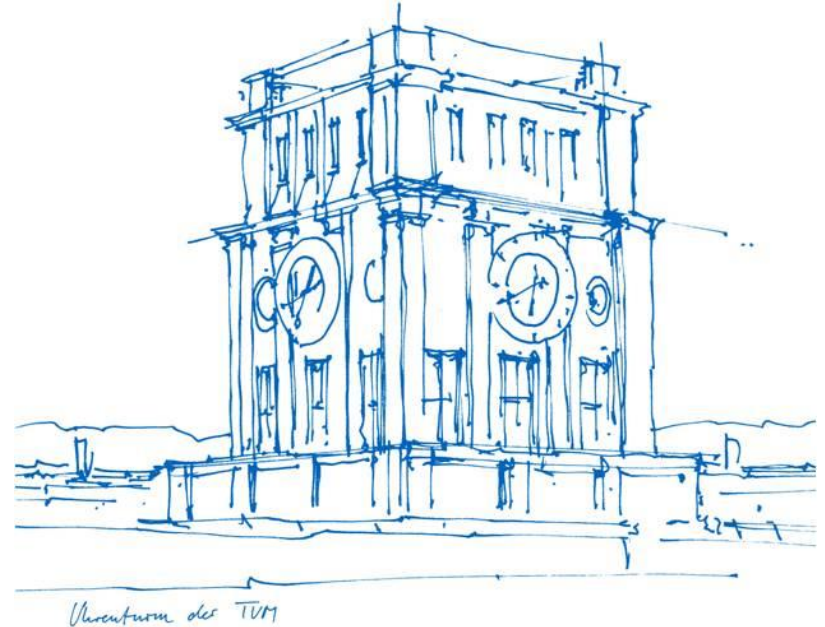
Examiner: Prof. Dr. David Wuttke

Technische Universität München

School of Management

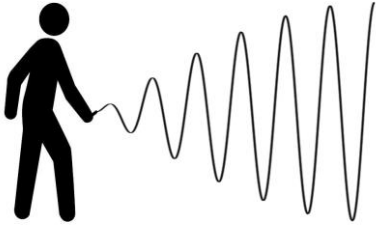
Campus Heilbronn

Heilbronn, 2026



Bullwhip Effect

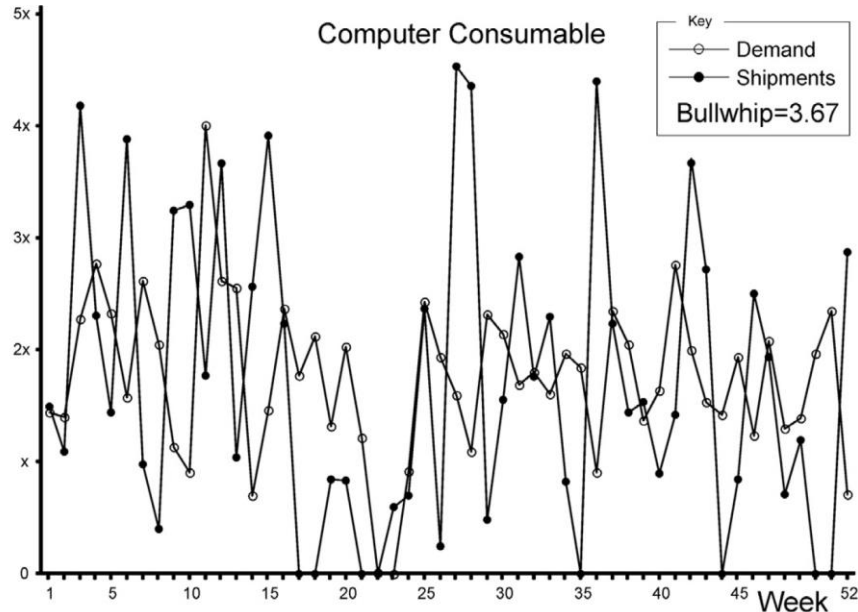
Bullwhip Effect



Bullwhip effect is prevalent if...

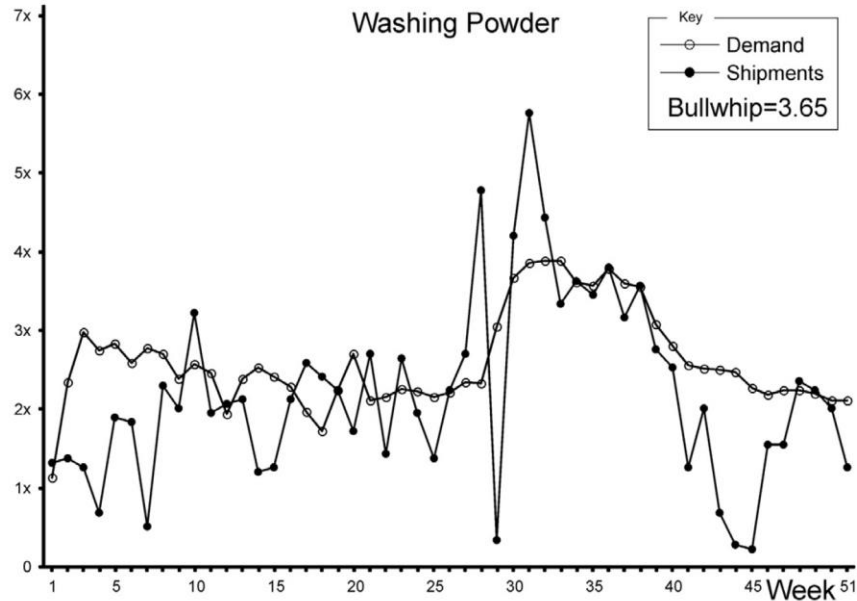
- the coefficient of variation of demand is higher at upstream stages of a supply chain.

Empirical Evidence pertaining to the Bullwhip Effect



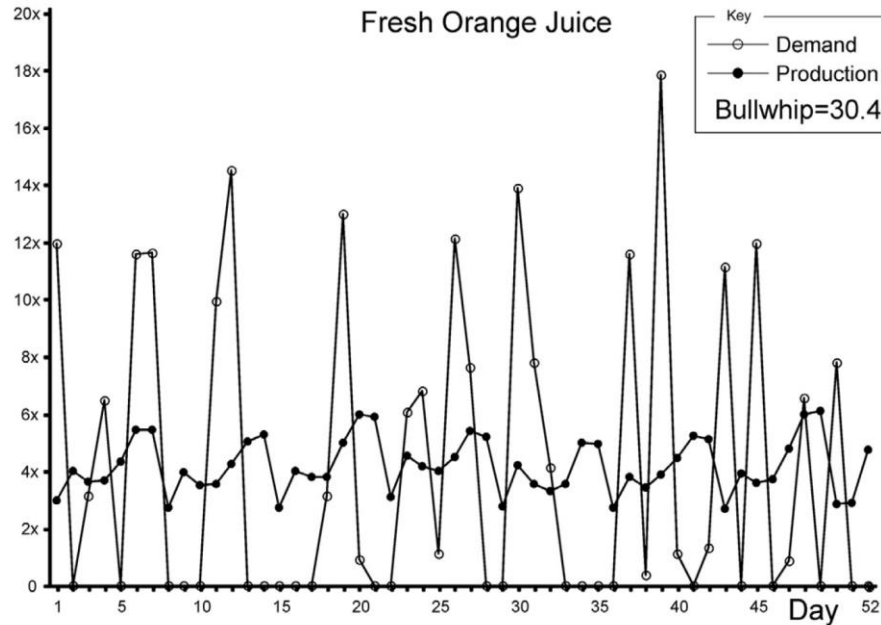
Source: Wang, X., & Disney, S. M. (2016). The bullwhip effect: Progress, trends and directions. *European Journal of Operational Research*, 250(3), 691-701.

Empirical Evidence pertaining to the Bullwhip Effect



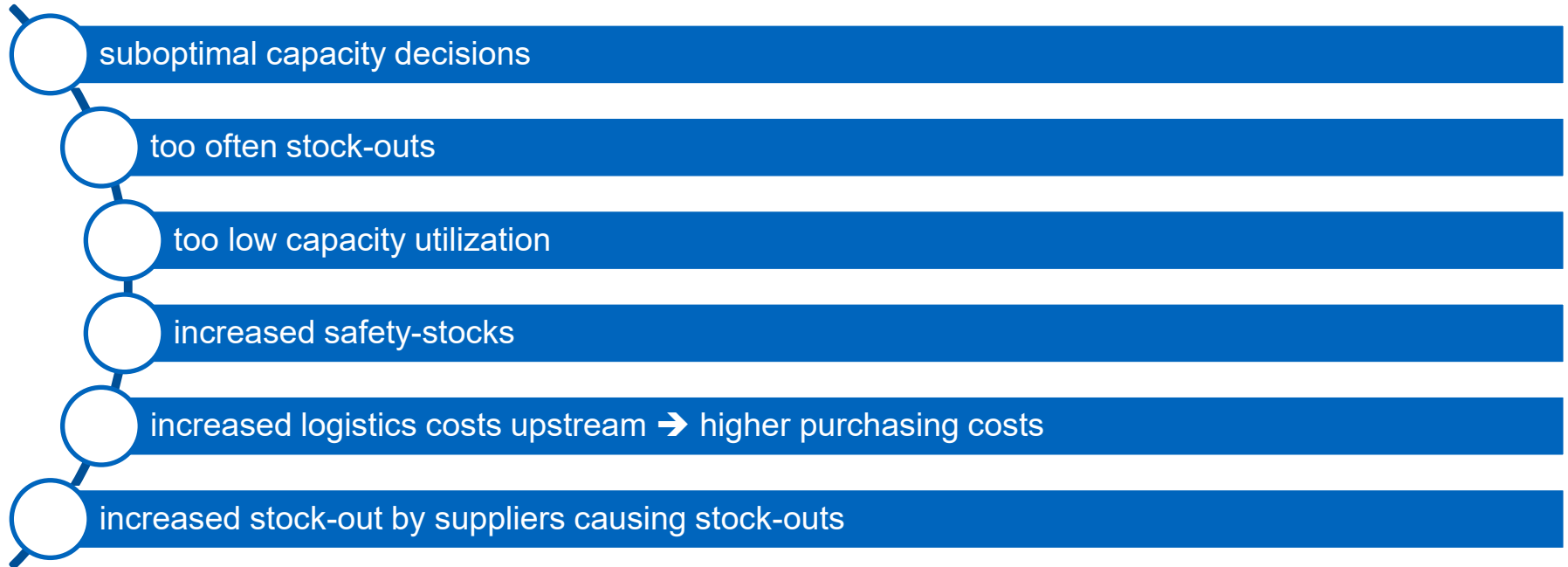
Source: Wang, X., & Disney, S. M. (2016). The bullwhip effect: Progress, trends and directions. *European Journal of Operational Research*, 250(3), 691-701.

Empirical Evidence pertaining to the Bullwhip Effect



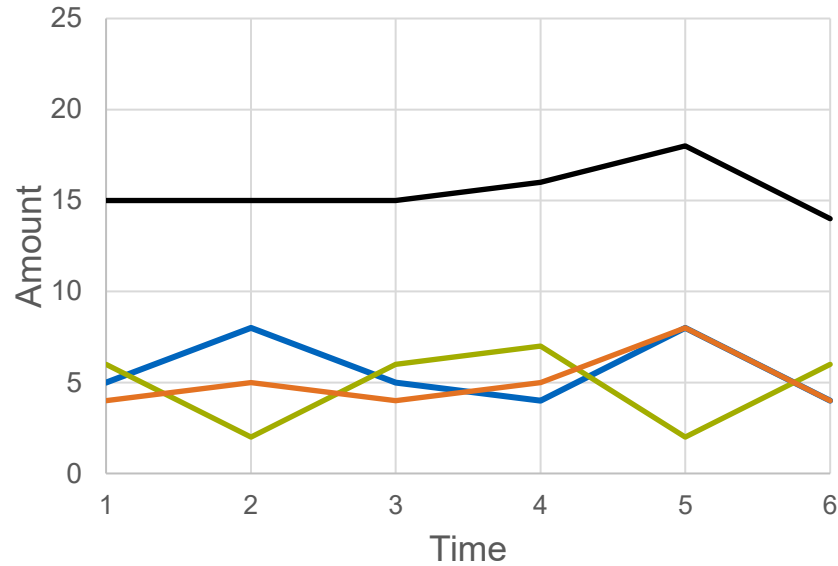
Source: Wang, X., & Disney, S. M. (2016). The bullwhip effect: Progress, trends and directions. *European Journal of Operational Research*, 250(3), 691-701.

Problems associated with the Bullwhip Effect

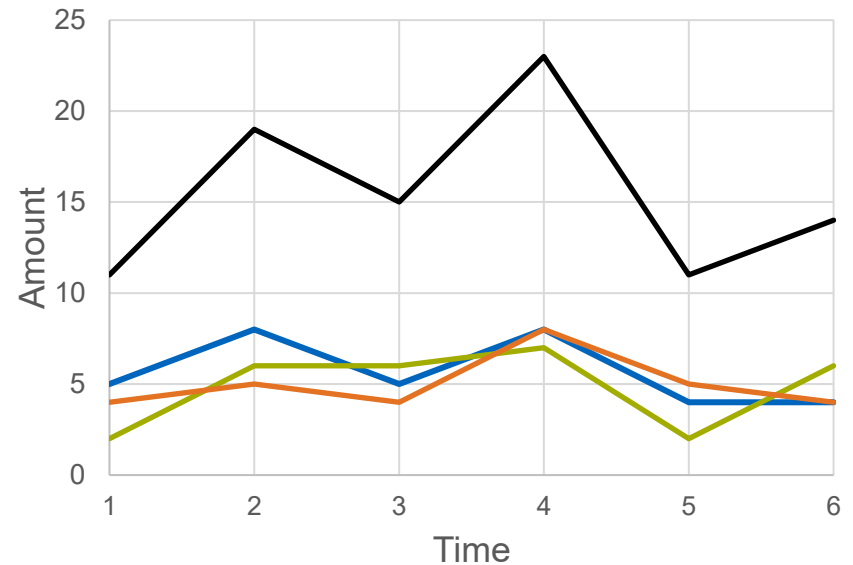


Causes of the Bullwhip Effect

Order Synchronization



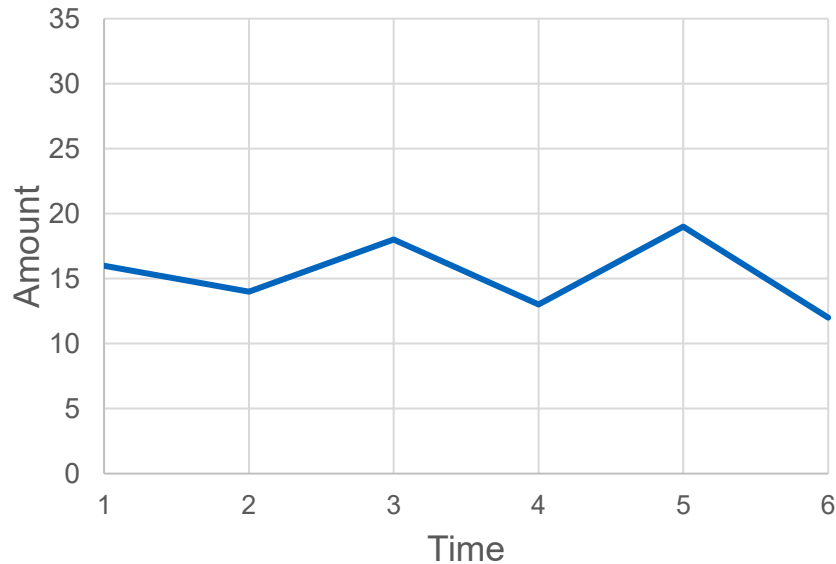
— Retailer 1 — Retailer 2 — Retailer 3 — Wholesaler



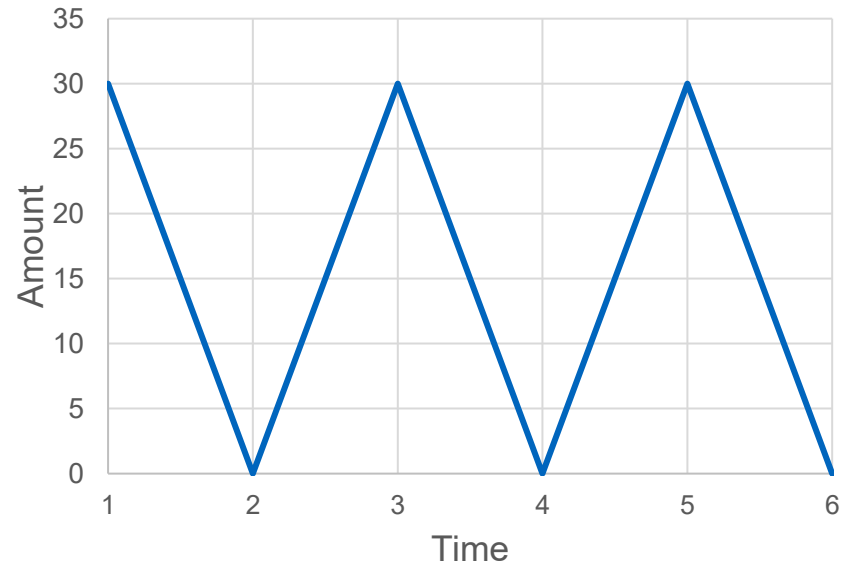
— Retailer 1 — Retailer 2 — Retailer 3 — Wholesaler

Order Batching

No batching (batch size = 1)

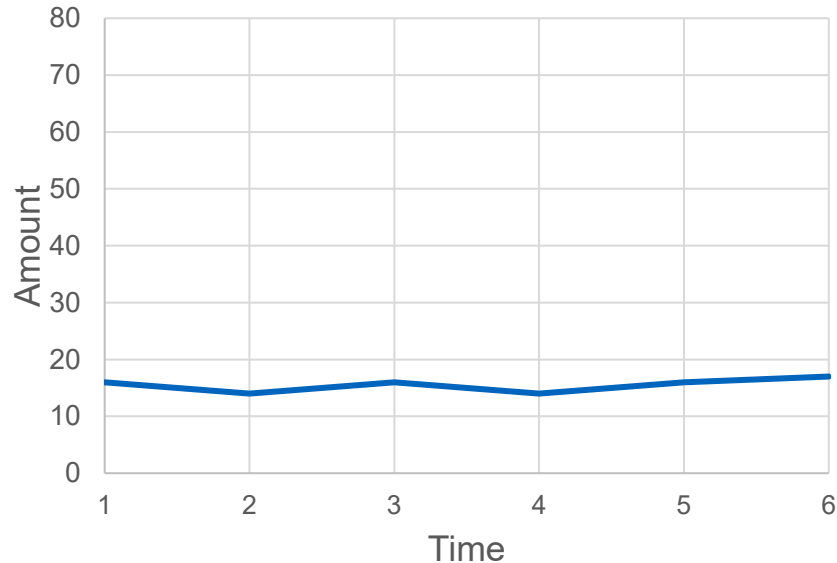


Batching (batch size = 15)

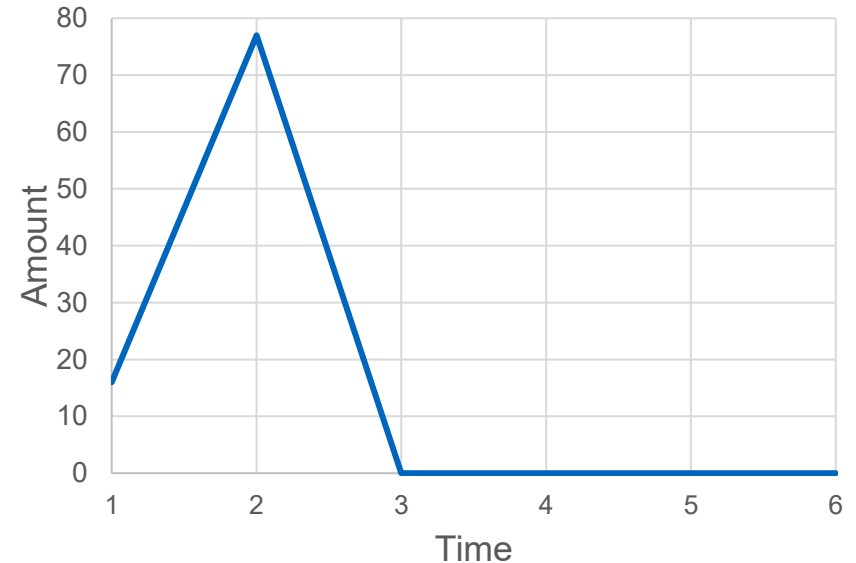


Trade Promotions and Forward Buying

Constant price (order = demand)



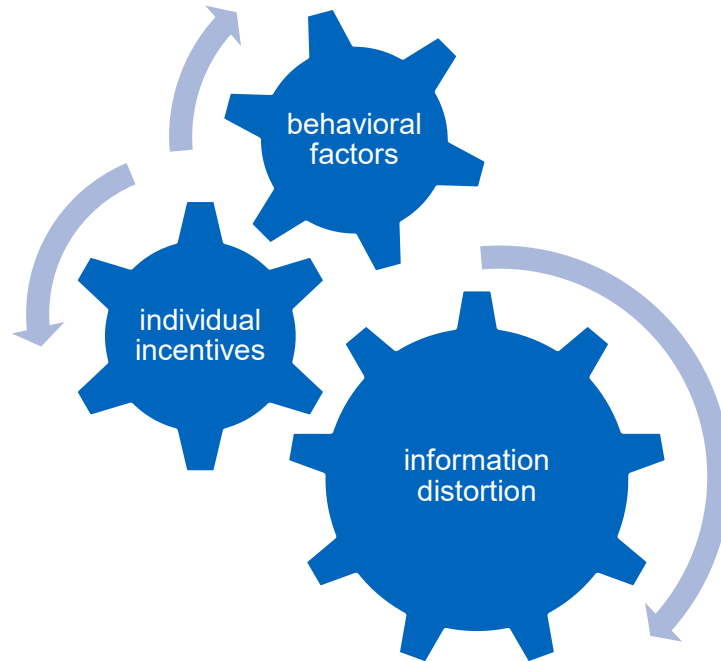
Discount in period 2



Shortage Gaming

Reactive and Over-reactive Ordering

Categorizing the Causes of the Bullwhip Effect



Mitigating the Bullwhip Effect

Sharing Information

POS

- points of sale data

EDI

- electronic data interchange

CPFR

- Collaborative planning forecasting and replenishment

Smooth Product Flow

VMI

- vendor managed inventory

EDLP

- every day low pricing

Lean Management

- value
- value stream
- flow
- pull
- perfection

Eliminate Pathological Incentives

Coordinate Sales Promotions

Restructure Return Policies

Turn-and-earn Policies